

Abhinandan Vijan

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Education

University at Buffalo, The State University of New York | NY, USA

Jan 2024-May 2025

Master of Science in Computer Science (GPA 3.56 / 4)

Coursework: Data Structure and Algorithms, Machine Learning, Deep Learning, Operating Systems, Data Model and Query Language

Thapar Institute of Engineering and Technology | Patiala, India

Aug 2016-June 2020

Bachelor of Engineering in Electrical Engineering (CGPA 8.0 / 10)

Experience

Software Developer | Buffalo, New York

Aug 2024-Dec 2024

University at Buffalo (Client - Best Practical Solutions LLC)

Project: Request Tracker AI Extension - An open-source request tracking and workflow management system

- Engineered AI-powered extensions using OpenAI's API and prompt chaining, enhancing customer support, ticket management.
- Built ticket summarization, sentiment analysis and response suggestion plugins using **Perl**, JavaScript, integrated with CkEditor

Senior Software Engineer | Pune, India

April 2023-Dec 2023

HSBC Software Development India

- Led development of .NET based Windows application to automate backup/restore of system and user profile for **400K+ users**, slashing manual intervention by **90%**.
- Architected modular plug-and-play DLL architecture, streamlining release cycles and improving reusability across product lines.
- Engineered secure RESTful APIs, optimized backend architecture, reducing average response time by **40%** and boosting resiliency.
- Conducted SQL Server performance tuning, cutting query execution times by **70%** using indexes and stored procedures.
- Enhanced application monitoring using **Splunk**, reducing issue resolution turnaround by **50%**.
- Collaborated in **Agile sprints**, participated in daily standups, peer code reviews, and contributed to the CI/CD pipeline.
- Received **Pat on the Back** award in October 2023 for contributing to the project success.

Software Engineer | Pune, India

Sept 2020-March 2023

HSBC Software Development India

- Maintained and enhanced multiple enterprise .NET web and desktop applications following OOP principles and MVC design patterns.
- Drove full-stack development of tools in an Agile team, supporting **20+** internal business units.
- Constructed and maintained 50+ data pipelines with SSIS and visualized business KPIs via 500+ SSRS reports as the primary owner.
- Automated recurring manual ETL tasks, **saving over 120 hours/month** across teams.
- Integrated applications with LDAP/Active Directory for secure access and wrote PowerShell scripts for application automation.

Technical Skills

Programming Languages: C#, Java, Python, JavaScript, C/C++, SQL, HTML, CSS

Frameworks: .NET, .NET Core, ASP.NET, MVC, Entity Framework, WPF, React.js

Databases: SQL Server (SSMS), PostgreSQL, MySQL

Tools & Platforms: AWS, Docker, GitHub, Postman, Splunk, SSIS, SSRS, PowerShell, Jira, Confluence, Visual Studio

Projects

PixeLens – An AI powered Image Captioning and Semantic Search Platform

Jan 2025-April 2025

- Created a web application that generates natural language captions from user-uploaded images using **Vision Transformers**.
- Designed a lightweight searchable database (SQLite) to store images, captions, and embeddings for fast retrieval and filtering.
- Engineered real-time text to image and image to image search using Vision Transformer embeddings and cosine similarity.

Splitwise App – A Full-Stack Expense Sharing Application

March 2025

- Built a full-stack application using **FastAPI (Python)** for backend APIs and **React.js** with Tailwind CSS for frontend UI
- RESTful APIs with secure user authentication using **JWT based token authentication** and added **Pydantic validations**.
- Designed and executed the logic to manage groups, track expenses and perform dynamic settlement using SQL Alchemy ORM

Sentiment Analysis on IMDB Movie Reviews - A deep learning solution to analyze customer feedback

June 2024

- Performed sentiment analysis on the IMDB movie review dataset using deep learning techniques.
- Achieved an accuracy of **90%** by fine-tuning CNN model for accurate sentiment classification.